Activity: Solve this

When two plants (A and B) belonging to the same genus but different species are crossed, the F_1 hybrid is viable and has more ornate flowers. Unfortunately, this hybrid is sterile and can only be propagated by vegetative cuttings. Explain the sterility of the hybrid and what would have to occur for the sterility of this hybrid to be reversed.



Plants are sessile. How do they protect themselves from the exposure of sunlight throughout the day?

Plants have effective DNA repair mechanism to prevent UV damage from sunlight. They produce an enzyme called photolyase, which can repair the thymine dimers and restore the structure of DNA.

SUMMARY

Chromosomal theory of inheritance states that Mendelian factors have specific loci on chromosomes and they carry information from one generation to the next generation. Genes located close together on the same chromosome and inherited together are called linked genes the phenomenon is called Linkage. Two types of linkage are complete linkage and incomplete linkage. The groups of linearly arranged linked genes are called Linkage groups. Crossing over is a biological process that produces new combination of genes by inter-changing the corresponding segments between non-sister chromatids of homologous pair of chromosomes. In this segments of DNA are broken and recombined to produce new combinations of alleles a process is called Recombination. The diagrammatic representation of distances between the adjacent genes which is directly proportional to the frequency of recombination between them is called genetic mapping. When any of the three or more allelic forms of a gene occupy the same locus in a given pair of homologous chromosomes, they are said to be multiple alleles. They are m, M1 and M2 of a single gene. Mutational events that take place within individual genes are called gene mutations or point mutation, whereas the changes that occur in structure and number of chromosomes are called chromosomal mutation. The agents which are responsible for mutation is called mutagens.

DNA metabolism includes replication, repair and recombination. Protein synthesis in eukaryotes is unique due to the capping, tailing and splicing. Transcription takes place in nucleus and translation in cytoplasm. AUG codes for methionine and it is monocistronic. Alternative splicing is a mechanism to overcome stress in plants. RNA editing takes place in chloroplast and mitochondria of plants which is of phylogenetic importance. Controlling elements gave a major shift in the gene organisation in plants to prove that DNA is not static but plastic entities. The first genome sequenced is *Arabidopsis thaliana*, which is a potential genetic tool to study the development and metabolism in plants.

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